

Credit Card Fraud Detection

A Machine Learning Approach to Financial Transaction Security

Step 1: Problem Understanding & Framing

Finance Domain | Binary Classification | June 2026

1. Business Context

Credit card fraud is ultimately a business loss problem before it is a technical one. When a transaction is fraudulent, the sale behind it is not truly collectible, the cardholder never authorized it, so revenue that appeared earned has to be reversed. Worse, a contested transaction doesn't resolve itself quietly: it triggers an investigation that consumes staff time and operating capital, even though the customer did nothing wrong. The institution ends up absorbing both the lost revenue and the cost of proving it.

This is compounded by how rare fraud actually is relative to legitimate activity. Fraudulent transactions typically make up a small fraction of one percent of all transactions, but each one is disproportionately costly once investigation, chargeback, and reputational effects are factored in. This combination of rarity and cost is precisely what makes fraud detection a hard data science problem: a model can't simply optimize for overall accuracy, since predicting "legitimate" every time would already be right almost all the time, and still useless.

This is not a marginal problem globally or locally. According to the Nilson Report, worldwide payment card fraud losses reached USD 33.45 billion in 2022, with losses projected to climb to roughly USD 41 billion by 2030 as digital transaction volume continues to grow [1]. In the Philippines, the shift toward digital wallets such as GCash and Maya alongside traditional credit cards, across a population now exceeding 115 million [2] has expanded the same exposure locally. The Bangko Sentral ng Pilipinas (BSP) has taken direct regulatory action on this front: BSP Circular No. 1140 (2022) mandated that supervised banks and financial institutions adopt fraud management systems [3], and BSP data shows supervised institutions reported PHP 5.82 billion in cyber-incident losses in 2024, up from PHP 5.67 billion in 2023, driven mainly by phishing, card-not-present fraud, and account takeovers [4].

Rule-based fraud detection, which relies on fixed thresholds and known fraud signatures, struggles against this because it can only catch what it has already seen. A new fraud pattern that falls outside existing rules passes through undetected, and fraudsters actively design their methods to stay under those thresholds. Machine learning offers a different approach: rather than checking transactions against a fixed

rulebook, it learns what normal behavior looks like and flags meaningful deviations from it, including patterns that have never been seen before.

It's worth noting a limitation up front: this approach is built to catch unauthorized transactions, fraud the cardholder didn't approve. It is not designed to catch social engineering scams, which have become increasingly common in the Philippines (e.g. phishing calls or fake bank messages that trick a customer into approving a transfer themselves). In those cases, the transaction is technically authorized by the legitimate cardholder, so it can look identical to normal behavior from the model's perspective, the failure happens at the human layer, not the transaction layer. Addressing that class of fraud requires a different set of controls (authentication, customer education) rather than transaction-pattern detection alone.

This project uses a European cardholder dataset, since Philippine transaction data is challenging to acquire for a project at this scale, it's held by individual institutions and not released publicly. The dataset itself won't transfer to the Philippine market, but the methodology will: how imbalance is handled, how thresholds are calibrated, and how model decisions are explained are lessons that apply regardless of geography, and can be adapted once local data becomes accessible.

2. Problem Statement

Core Problem

Given a dataset of European cardholder transactions, build a machine learning model that can accurately distinguish between fraudulent and legitimate transactions in real time, while minimizing both false positives (legitimate transactions incorrectly blocked) and false negatives (fraud transactions that go undetected).

These two error types don't carry equal weight, and both have a real business consequence, not just a technical one. A false negative means fraud goes through: the sale is uncollectible, and the institution absorbs the loss plus the cost of investigating the dispute, as outlined above. A false positive means a legitimate transaction gets declined and from the customer's point of view, there's no way to know that a fraud model made that call. What they experience is simply their card getting rejected. That friction gets attributed to the bank itself, not to an invisible detection system working in the background, and repeated false declines are a real driver of customers switching to a competitor. Minimizing false positives isn't just a precision metric on paper, it's protecting the relationship between the institution and the customer.

The challenge is compounded by extreme class imbalance: in real-world fraud datasets, fraudulent transactions typically represent less than 0.2% of all records. This imbalance means a naive model that classifies every transaction as legitimate would score 99.8% accuracy while being completely useless from a business standpoint, since it would catch zero fraud.

This project addresses the imbalance problem through appropriate resampling techniques (SMOTE, undersampling), threshold tuning, and evaluation metrics that are robust to class imbalance (AUC-ROC, Precision-Recall AUC, F1-Score) rather than relying on accuracy alone.

3. Data Science Task Definition

Attribute	Definition
Task Type	Supervised Binary Classification
Target Variable	Class (0 = Legitimate, 1 = Fraudulent)
Input Features	28 PCA-derived features (V1–V28), Time, Amount
Dataset	Kaggle: Credit Card Fraud Detection (ULB, 2013)
Records	284,807 transactions 492 fraud (0.173%)
Train/Test Split	80% training / 20% testing (stratified)

4. Success Metrics

4.1 Primary ML Metrics

Metric	Target	Rationale
AUC-ROC	≥ 0.95	Primary metric; robust to class imbalance; measures overall discriminatory power
Precision-Recall AUC	≥ 0.80	More informative than ROC when positive class is rare
F1-Score (Fraud Class)	≥ 0.80	Harmonic mean of precision and recall for the minority class
Recall (Fraud Class)	≥ 0.85	Prioritise catching fraud; missed fraud is costlier than false alarms
Precision (Fraud Class)	≥ 0.75	Minimise customer friction from false declines

4.2 Business KPIs

KPI	Description
Fraud Detection Rate (FDR)	% of actual fraud transactions correctly identified by the model
False Positive Rate (FPR)	% of legitimate transactions incorrectly blocked (customer friction proxy)
Estimated Annual Savings	Model-detected fraud volume $\times$ average fraud transaction value
Cost-per-False-Positive	Estimated cost of manual review per false alert triggered

KPI	Description
Fraud Loss Reduction (%)	Reduction in total fraud losses vs. baseline (no model)

Note: The cost asymmetry of fraud detection (a missed fraud is typically 5–10x more costly than a false positive) informs the decision to prioritise recall over precision when selecting the final model threshold.

5. Stakeholder Mapping

Stakeholder	Primary Concern	Metric That Matters
Risk & Fraud Operations Team	Catch as much fraud as possible	Recall, FDR
Customer Experience Team	Minimize false declines on good customers	Precision, FPR
Finance / CFO	Reduce total fraud losses and operational costs	Estimated Savings, ROI
Compliance & Regulatory	Audit trails, explainability, bias controls	SHAP explanations, fairness audits
ML Engineering Team	Model performance, latency, maintainability	AUC-ROC, F1, inference time

6. Project Scope & Constraints

6.1 In Scope

- End-to-end ML pipeline: EDA, preprocessing, modelling, evaluation, and explainability
- Four classification models spanning a range of complexity — Logistic Regression, Decision Tree, Random Forest, and XGBoost. Evaluated across multiple resampling strategies
- Handling class imbalance via SMOTE, undersampling, and threshold optimization
- Model explainability via SHAP values
- Bias & fairness audit across transaction amount segments
- Two slide decks: technical and business-facing
- GitHub repository with reproducible code

6.2 Out of Scope

- Real-time deployment infrastructure (addressed optionally in Step 8)
- Integration with live payment systems or card networks
- Collection of new or proprietary transaction data

6.3 Assumptions

- The Kaggle dataset is representative of European cardholder behaviour (2013)
- PCA-transformed features (V1–V28) preserve sufficient signal for fraud detection
- All model development is performed in Google Colab using Python 3.x
- The cost of a missed fraud is assumed to be 5x the cost of a false positive

7. Expected Business Impact

Illustrative ROI Scenario

Assume a bank processes 1,000,000 transactions per month. At a 0.173% fraud rate (consistent with the dataset used in this project), approximately 1,730 transactions are fraudulent. Using an average fraud value of approximately PHP 102,600 — derived from BSP-reported cyber fraud data (card-not-present fraud, phishing, and account takeover incidents totaling PHP 1.5 billion across 14,610 cases in 2022) total monthly fraud exposure is approximately PHP 177.5 million. A model achieving 85% recall would detect and prevent approximately PHP 150.9 million of this exposure monthly, translating to a potential annual savings of roughly PHP 1.81 billion, offset by investigation costs associated with false positives. [5]

Note: This figure spans multiple BSP-reported cyber fraud categories, not credit card transactions exclusively, since it is the most granular publicly available breakdown. It should be read as an order-of-magnitude illustration rather than a precise industry benchmark.

Beyond direct financial savings, a well-calibrated fraud detection model delivers secondary benefits: improved customer trust, reduced manual review workloads, faster transaction decisioning, and a compliance-ready audit trail and all of which contribute to long-term institutional value.

References

- [1] The Nilson Report. (2025, January 6; 2026, January 7). Payment Card Fraud Losses Approach \$34 Billion / Global Card Fraud Losses at \$33 Billion. Santa Barbara, CA: The Nilson Report.
- [2] Worldometer / StatisticsTimes. (2026). Philippines Population (2026). Based on UN Department of Economic and Social Affairs, Population Division estimates.
- [3] Bangko Sentral ng Pilipinas. (2022). Circular No. 1140, Series of 2022: Amendments to Regulations on Information Technology Risk Management.
- [4] Savyint. (2025). Philippines BSP Circular No. 1213 and Compliance Solutions for Financial Institutions, citing BSP-reported cyber incident losses, 2023–2024.
- [5] Inquirer.net Business. (2024, January 23). Card fraud most rampant cybercrime in PH, citing Bangko Sentral ng Pilipinas cyber incident data for 2022.